Vidyalankar Institute of Technology
Semester V – INFT Mid Semester Assessment

Date: 18th Sept 2024 | Data Warehouse and Mining | 20 Marks/ 1 hour

1 Solve any two (5 marks each) CO

A Explain different types of attributes with examples CO4

B Suppose the data for analysis includes the attribute salary. The salary values for the data tuples are (in increasing order): 25000, 30000, 32000, 35000, 40000, 42000, 45000, 47000, 50000, 55000, 58000, 60000, 62000, 65000, 70000, 75000, 80000, 85000, 90000, 95000, 100000, 105000, 110000, 115000, 120000, 130000, 140000. CO4

i. Apply Min-Max normalization to transform the salary value of 75000 within the range [0.0, 1.0]
ii. Apply z-score normalization to transform the salary value of 75000, where the standard deviation for salary is 20000.

C Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8) CO4

i. Compute the Euclidean distance between the two objects.
ii. Compute the Manhattan distance between the two objects.
iii. Compute the Supremum distance between the two objects.

2 Solve any one (10 marks each)

A Suppose that the data for analysis include the attribute age.The age values for the data tuples are (in increasing order) as: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 CO4

i. What is the mean of the data? What is the median?
ii. What is the mode of the data? Comment on the data's modality(i.e. bimodal, trimodal, etc.)
iii. What is the midrange of the data
iv. Find the first Quartile (Q1) and the third Quartile (Q3) of the data.
v. Give the five number summary of the data.
vi. Find the Outlier if any.
vii. Show a boxplot of the data.

B Following is the training set given for play tennis example. CO6

	Age	income	student	credit_rating	buys_computer
1	<=30	High	no	fair	no
2	<=30	High	no	excellent	no
3	31...40	High	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	Low	yes	fair	yes
6	>40	Low	yes	excellent	no
7	31...40	Low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	Low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31...40	medium	no	excellent	yes
13	31...40	High	yes	fair	yes
14	>40	medium	no	excellent	no

Find out the class label for the data to be classified by using Naïve Bayesian:

X = (age <=30, Income = medium,Student = yesCredit_rating = Fair)

CO1	Demonstrate a comprehensive understanding of the fundamental concepts of data warehousing and its architecture.
CO2	Design a data warehouse schema using dimensional modelling techniques and explain the ETL process involved in data warehousing.
CO3	To introduce concepts and fundamentals of data lakes
CO4	Understand data mining principles and perform data preprocessing and Visualization.
CO5	Understand the concept of data mining and identify appropriate data mining algorithms to solve real-world problems.
CO6	Implement basic data mining algorithms such as classification, clustering, and association mining

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Question-1(a)

Attributes describes the characteristics of data points and are categorized based on their nature and kind of values they hold. Summary of different types of attributes are as follows :-

(1) **Nominal Attributes:-** Nominal attributes are categorical attributes where values represent distinct categories without any intrinsic ordering or ranking.

Example :- Gender (Male, Female)
Color (red, blue, green)

(2) **Ordinal Attributes:-** Ordinal attributes represent categories with a meaningful order or ranking among them but with unspecified distances between categories.

Example :- Educational Level (High school, bachelor's, master's, Ph.D.)

(3) **Interval Attributes:-** Interval attributes have ordered values with consistent and meaningful difference.

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		b/w values, but lack of a true zero point.
		Example :- Temperature in Celsius
	④	Ratio Attributes :- Ratio attributes are similar to internal attributes but include a true zero point, indicating the absence of quantity being measured.
		Example :- Salary
	⑤	Binary Attributes :- Binary attributes have only two possible values, often representing the presence or absence of a characteristics.
		Example : Has a car (Yes or No)

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Question - 1(b)

(1) **Min - Max Normalization :-**

$$x' = \frac{x - min}{max - min}$$

$x = 75000$
$min = 25000$
$max = 140000$

$$x' = \frac{75000 - 25000}{140000 - 25000}$$

$$x' = \frac{50000}{115000}$$

$$\boxed{x' \approx 0.4348}$$

(2) **Z - Score Normalization :-**

$$Z = \frac{x - \mu}{\sigma}$$

μ = mean of dataset

Sum of Salaries = 1,597,000

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Number of values = 27

$$\mu = \frac{1597000}{27}$$

$$\approx 59,037$$

$$Z = \frac{75000 - 59037}{20000}$$

$$= \frac{15963}{20000}$$

$$\boxed{Z \approx 0.798}$$

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Question-1(c)

$A = (22, 1, 42, 10)$

$B = (20, 0, 36, 8)$

① Euclidean Distance :-

$$d_{Euclidean} = \sqrt{\sum_{i=1}^{n} (A_i - B_i)^2}$$

$$= \sqrt{(22-20)^2 + (1-0)^2 + (42-36)^2 + (10-8)^2}$$

$$= \sqrt{4+1+36+4}$$

$$= \sqrt{45}$$

$$\approx \boxed{6.708}$$

② Manhattan Distance :-

$$d_{Manhattan} = \sum_{i=1}^{n} |A_i - B_i|$$

$$= |22-20| + |1-0| + |42-36| + |10-8|$$

$$= 2+1+6+2$$

$$= \boxed{11}$$

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	(3)	Supremum Distance (or Chebyshev) –

$$d_{supremum} = \max_i |A_i - B_i|$$

$$d_{supremum} = \max(|22-20|, |1-0|, |42-36|, |10-8|)$$

$$= \max(2, 1, 6, 2)$$

$$= \boxed{6}$$

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Question-2(a)

Data =

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 52, 70

① Mean :-

$$\text{Mean} = \frac{\Sigma (\text{All values})}{\text{Number of values}}$$

$$= \frac{282}{27} \approx \boxed{21.56}$$

② Median :-

Median position = 14

$$\boxed{\text{Median} = 25}$$

③ Mode :- $\boxed{\text{Mode} = 25}$ (Unimodal)

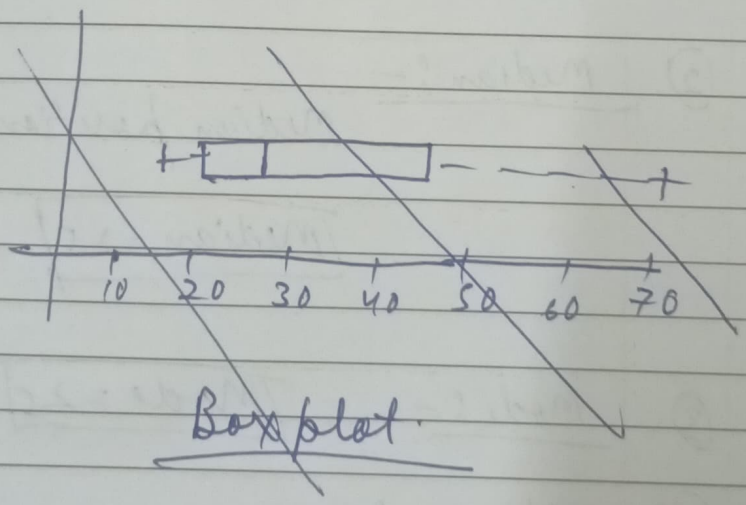
④ Midrange :-

$$\text{Midrange} = \frac{\text{min value} + \text{max value}}{2}$$

$$= \frac{13 + 70}{2} = \boxed{41.5}$$

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	(4)	$Q_1 = 20$ (7th value)
		$Q_3 = 35$ (7th from end)
	(5)	Five number Summary :-
		min = 13
		$Q_1 = 20$
		Median = 25
		$Q_3 = 35$
		max = 70



Boxplot

	(6)	Outliers :-

$IQR = Q_3 - Q_1 = 35 - 20 = 15$

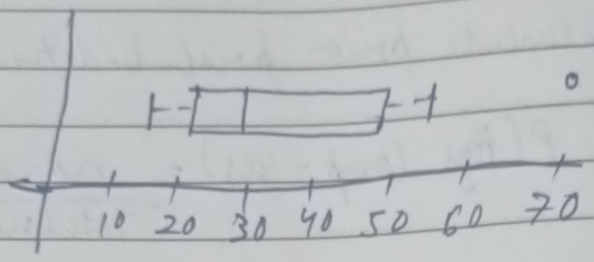
Lower Bound : $Q_1 - 1.5 \times IQR = -2.5$

Upper Bound : $Q_3 + 1.5 \times IQR = 57.5$

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Outliers : 70



Total Marks of Question no.		Examiner 1	

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Question-2 (b)

The step by step process is :-

① Calculate prior probabilities :-

$$P(\text{Buys_Comp} = \text{yes}) = \frac{\text{no. of Yes}}{\text{Total no. of Instances}} = \frac{9}{14}$$

$$P(\text{buys_comp} = \text{no}) = \frac{5}{14}$$

② Calculate Conditional Probabilities :-

For "yes" class:

→ $P(\text{age} \leq 30 \mid \text{Buys_comp} = \text{yes})$

→ Instance with age ≤ 30 and "yes" : 3
→ Total "yes" instance : 9

$$P(\text{age} \leq 30 \mid \text{buys_comp} = \text{yes}) = \frac{3}{9} = \frac{1}{3}$$

→ $P(\text{income} = \text{medium} \mid \text{buys_computer} = \text{yes})$

$= 4/9$

→ $P(\text{student} = \text{yes} \mid \text{buys_comp} = \text{yes})$

$= 6/9$

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→ P(credit_rating = fair / buys_comp = yes)

= 6/9 = 2/3

For "no" class :-

→ P(age ≤ 30 / buys_comp = no)

= 3/5

→ P(income = medium / buys_comp = no)

= 2/5

→ P(student = yes / buys_comp = no)

= 3/5

→ P(credit_rating = fair / buys_comp = no)

= 3/5

(3) Apply Bayes' Theorm:

→ To classify X, we use:

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$P(\text{buys_comp} = \text{yes} \mid \text{features}) \propto$

$P(\text{age} \leq 30 \mid \text{buys_comp} = \text{yes}) \times$
$P(\text{income} = \text{medium} \mid \text{buys_comp} = \text{yes}) \times$
$P(\text{students} = \text{yes} \mid \text{buys_comp} = \text{yes}) \times$
$P(\text{credit_rating} = \text{fair} \mid \text{buys_comp} = \text{yes}) \times$
$P(\text{buys_comp} = \text{yes})$

$$\frac{1}{3} \times \frac{4}{9} \times \frac{5}{9} \times \frac{2}{3} \times \frac{9}{14} = \frac{360}{2835}$$

→ Similarity

$P(\text{buys_comp} = \text{no} \mid \text{features}) \propto$

$$\frac{3}{5} \times \frac{2}{5} \times \frac{3}{5} \times \frac{3}{5} \times \frac{5}{14} = \frac{270}{4375}$$

(4) Normalize Probabilities :-

$$\text{Ratio} = \frac{\frac{360}{2835}}{\frac{270}{4375}} \approx \boxed{1.09}$$